# Error Budgets

*the failure you're allowed, and how to spend it*

Watch the 4-min video

**In one line:** the error budget is 1 − SLO, the amount of failure the SLO allows; you alert on how fast you burn it, and a pre-agreed policy says what happens when it runs out: ship freely while budget remains, freeze launches and fix reliability when it's gone.

## 1 Ship faster or slow down?

Product wants more releases. The reliability team wants fewer, because the last release broke checkout. Both argue from gut feeling. An **error budget** replaces the argument with a number both sides agreed on before the incident: the SLO from lesson 2 already says how much failure is acceptable.

## 2 The budget and the burn rate

```
budget = 1 − SLO    ·    burn rate = observed error rate ÷ (1 − SLO)
```

With a **99.9%** SLO the budget is **0.1%**: about **43 minutes** of total downtime in 30 days (0.001 × 43,200 min), or 1 million failed requests out of a billion. The budget is meant to be **spent**: on releases, experiments, migrations and planned maintenance.

**Burn rate** says how fast you're spending it. At 1× you use exactly the whole budget over the window. If 1% of requests fail against a 99.9% SLO, the burn rate is 0.01 ÷ 0.001 = **10×**, and a 30-day budget is gone in 3 days.




**Figure 1.** Burn rate is the slope. The same error rate is harmless or urgent depending on how quickly it empties the budget.

## 3 Alert on burn rate, with two windows

Raw error-rate alerts are noisy: 2% errors for one minute doesn't matter, 0.2% all week does. Alert instead on **burn rate**, using a long window to make sure the burn is significant and a short window (about 1/12 of the long one) so the alert stops soon after the problem is fixed. The Google SRE Workbook's recommended starting point for a 30-day, 99.9% SLO:

| Action | Long window | Short window | Burn rate | Budget used when it fires |
|---|---|---|---|---|
| Page | 1 hour | 5 minutes | 14.4× | 2% |
| Page | 6 hours | 30 minutes | 6× | 5% |
| Ticket | 3 days | 6 hours | 1× | 10% |

The video rounds this to "page at 14× over 1 hour, ticket at 3× over 6 hours"; the exact thresholds are a choice, the shape is what matters: **fast burns page a human, slow burns open a ticket**. Both conditions (long *and* short window) must be true to fire.

## 4 The error budget policy

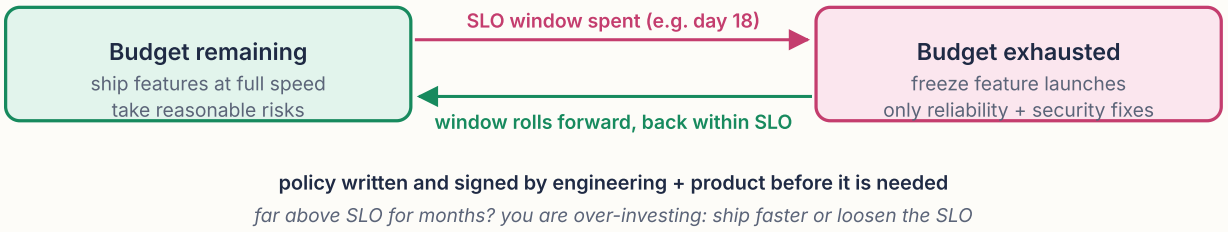



**Figure 2.** The policy is agreed in advance, so when the budget runs out the data decides, not the loudest voice.

- **Budget left:** teams release at normal speed and can take risks such as a big migration.
- **Budget spent:** feature launches freeze; only reliability fixes and security patches ship until the rolling window recovers.
- **Way above SLO:** you're over-investing in reliability, so release faster or spend the spare budget on riskier work.
- **Postmortems** are triggered by, and prioritised by, how much budget an incident consumed (lesson 70).

**WORKED EXAMPLE: A LAUNCH FREEZE ON DAY 18**

Checkout has a **99.9% SLO over 30 days**, so about 43 minutes of budget. A bad deploy causes 25 minutes of errors on day 9; a database failover adds 20 minutes on day 18. The budget is gone. As the policy says, feature launches stop and the team spends the rest of the window on **rollback automation** and **better canary checks** (lesson 75). As the bad days roll out of the 30-day window, budget returns and launches resume, with no meeting needed to decide it.

**COMMON MISTAKES**

- Writing the policy during the incident, when nobody can agree.
- Alerting on every error spike instead of burn rate, so on-call learns to ignore pages.
- Treating the budget as a target to avoid at all costs, instead of something to spend.
- No exception for security fixes in the freeze.

## 5 Key terms

**Error budget**: 1 − SLO: the failure the SLO allows in a window

**Burn rate**: error rate ÷ (1 − SLO); 1× empties it in exactly one window

**Multi-window alert**: fires only when a long and a short window both burn fast

**Fast burn**: high burn rate: page someone now

**Slow burn**: low but steady burn: open a ticket

**Budget policy**: pre-agreed actions when the budget is exhausted

## 6 Interview questions

**Q1. What is an error budget for a 99.95% SLO over 30 days?** BASIC

0.05% of requests, or about 21.6 minutes of full downtime (0.0005 × 43,200 minutes).

**Q2. What does a burn rate of 10 mean?** BASIC

You're failing ten times faster than the SLO allows; a 30-day budget would be exhausted in 3 days.

**Q3. How do error budgets change engineering decisions?** SCENARIO

With budget left, teams ship and take risks; when it's spent, launches freeze except reliability and security fixes, as the agreed policy says. Services far above their SLO can move faster.

**Q4. How should you alert on an SLO?** SCENARIO

On burn rate with multiple windows: page on fast burns (e.g. 14.4× over 1 h and 5 min), ticket on slow burns (e.g. 1× over 3 days, or 3× over 6 h).

**Q5. Why add a short window to a 1-hour burn-rate alert?** TRICKY

Without it the alert keeps firing for up to an hour after the fix, because the 1-hour average stays high; the short window makes it reset quickly.

**Q6. A service has used none of its budget for six months. Good news?** TRICKY

Not entirely: it suggests over-investment or an SLO that's too loose. Ship faster, run riskier experiments, or check that the SLI really reflects users.

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